
CLEAN FEATURE ANCHORING: MITIGATING THE ACCURACY-ROBUSTNESS TRADE-OFF BY SELF-DISTILLING A NATURALLY TRAINED MODEL

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ABSTRACT

Adversarial training significantly improves adversarial robustness, but robust models attain substantially lower standard accuracy than naturally trained ones. This accuracy gap has motivated a line of work that brings a naturally trained network into adversarial training. Prior methods commonly transfer its logits or combine representation guidance with a label loss. We also employ a naturally trained network, but we exploit it far more directly: the label loss is discarded entirely and the backbone is trained against the natural network alone. In this paper, we propose Clean Feature Anchoring (CFA), which anchors the adversarial feature of the student to the clean feature of the teacher rather than to its logits. Because one fixed target bounds both clean fidelity and local feature variation, the objective needs no coefficient balancing two losses and no temperature. We further assign the attack radius per sample according to the input sensitivity of the loss under a fixed total budget, approximately equalizing the first-order loss change across examples. Controlled experiments separate the anchor from the training recipe: replacing it by label cross-entropy costs 4.63 standard and 4.45 AutoAttack points under matched conditions, and giving ten baselines the same warm start and stack leaves all of them behind CFA. Particularly, CFA attains 64.75% standard accuracy at 27.98% AutoAttack accuracy on CIFAR-100 with the ResNet-18 architecture, and consistent improvements are observed on CIFAR-10 and Tiny-ImageNet.

1 INTRODUCTION

Adversarial training improves worst-case accuracy but often sacrifices clean accuracy (Madry et al., 2017; Tsipras et al., 2018; Zhang et al., 2019). Robust distillation can soften this trade-off, but requires an adversarially trained teacher (Goldblum et al., 2020; Zhu et al., 2021; Zi et al., 2021; Huang et al., 2023; Lee et al., 2025). A naturally trained teacher is cheaper and more accurate on clean inputs, yet vulnerable to attack. What should a robust student learn from such a teacher?

Prior methods use natural teachers’ predictions or representations (Cui et al., 2021; Suzuki et al., 2023; Cho & Kim, 2026; Zhao et al., 2024). We study a particularly simple answer: match the teacher’s *clean feature* throughout adversarial training, with the inherited classifier fixed. The same target drives attack generation and student learning, without a separate label loss or temperature search. This design exposes two choices beyond the target itself: which natural checkpoint to use, and how to allocate perturbations across examples.

Clean Feature Anchoring (CFA) makes the target explicit: the student backbone learns to match one clean teacher representation even at adversarial inputs. A fixed-anchor bound relates this loss to clean teacher fidelity and local feature variation; it does not rely on the teacher itself being robust.

The teacher checkpoint changes what the student learns. Across two independent CIFAR-100 training trajectories, later natural checkpoints shift students toward higher robustness and lower clean accuracy. Moreover, label-smoothed and mixup teachers with nearly identical clean accuracy pro-

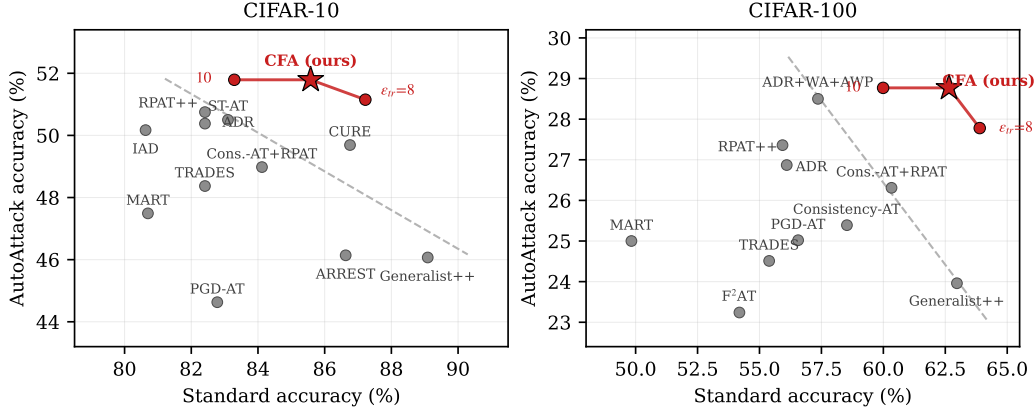


Figure 1: Clean–AutoAttack trade-off on CIFAR-10 and CIFAR-100 with ResNet-18 at $\epsilon = 8/255$. Grey points are published results (Table 29); the CFA curve varies the training radius over 100-epoch runs, and the star marks $\epsilon_{\text{train}} = 8.8/255$.

duce students 4.18 clean points apart. Teacher accuracy alone is therefore not a sufficient description of the supervision a checkpoint provides.

Even with the teacher fixed, a uniform pixel radius need not perturb the anchor loss equally across examples. We allocate a fixed batch budget inversely to the loss’s input sensitivity. In the final recipe, this changes clean accuracy from 62.40% to 64.75% while AA remains near 28%. A separate control shows that reassigning the same radii by sample difficulty weakens the result. The teacher determines the reference, while radius allocation determines where to enforce it.

CFA improves the measured trade-off across CIFAR-10, CIFAR-100, and Tiny-ImageNet. On CIFAR-100 with ResNet-18 it reaches 64.75% clean and 27.98% AA, exceeding ARREST by 1.58 AA points at comparable clean accuracy (Table 3). Target, recipe, and radius controls distinguish the anchor’s contribution from that of the training stack.

Our contributions are:

- Adversarial training driven by a natural teacher’s clean feature alone, with no label loss and no temperature, separated from the training recipe by objective, initialization, schedule and stack controls; one fixed target bounds clean fidelity and local feature variation together, which is why no balancing coefficient appears.
- Evidence that the teacher checkpoint sets the student’s operating point rather than its quality: two independent trajectories move clean accuracy and robustness in opposite directions, and two teachers of nearly equal clean accuracy produce students 4.18 clean points apart.
- A radius allocation read from the loss’s own input sensitivity, which holds at the reported configuration and survives controls that keep the budget, and even the multiset of radii, fixed.

2 RELATED WORK

Adversarial training and its targets. PGD adversarial training optimizes a model on worst-case inputs (Madry et al., 2017); TRADES separates clean prediction from local consistency (Zhang et al., 2019), and MART reweights misclassified examples (Wang et al., 2020). Subsequent work improves generalization through label smoothing, consistency regularization, weight averaging, and adversarial weight perturbation (Pang et al., 2020; Tack et al., 2022; Rice et al., 2020; Wu et al., 2020). ADR replaces one-hot labels with targets produced by an annealed momentum teacher (Wu et al., 2024). These approaches motivate our study of how the supervisory target affects the clean–robust trade-off.

Adversarial distillation. ARD transfers a robust teacher’s predictions during adversarial training (Goldblum et al., 2020). RSLAD emphasizes robust soft labels (Zi et al., 2021), IAD discounts unreliable teacher guidance (Zhu et al., 2021), and AdaAD uses the teacher in both adversarial-example generation and student optimization (Huang et al., 2023). IGDM adds the teacher’s local input geometry to logit-based distillation (Lee et al., 2025). These methods are designed to inherit robustness from an adversarially trained teacher. In contrast, our teacher is trained only on clean data, and its clean representation remains fixed during the student’s inner maximization.

Guidance from naturally trained models. BGAT regresses adversarial logits to a fixed natural teacher’s clean logits, and LBGAT extends this to a jointly trained natural branch (Cui et al., 2021). ARREST starts from a naturally pretrained model and regularizes latent representations while retaining adversarial cross-entropy and noisy replay (Suzuki et al., 2023). HAT uses a natural helper to label examples outside the training ball (Rade & Moosavi-Dezfooli, 2022), DP-FAT distills direction-preserving logits (Cho & Kim, 2026), and B-MTARD combines clean and robust teachers with adaptive temperature and loss balancing (Zhao et al., 2024). CFA uses a fixed clean feature target for both attack generation and backbone learning, with an inherited classifier. We study how the natural-teacher checkpoint and sensitivity-based radius allocation affect transfer under this objective.

Sample-dependent perturbation budgets. IAAT and CAT adapt the training radius to sample difficulty, while MMA directly increases an input-space margin (Balaji et al., 2019; Cheng et al., 2022; Ding et al., 2020). Our allocation has a different source: it redistributes a fixed batch budget according to the input sensitivity of the loss being optimized. The comparison is therefore not between adaptive and uniform radii alone, but between the signals used to assign the same budget.

3 ANALYSIS

A natural teacher offers accurate clean predictions, but these need not be the most useful targets for adversarial training. We compare predictive and feature targets, establish what a fixed clean target controls, and then ask how the teacher checkpoint changes the student’s behavior.

3.1 PRELIMINARIES

Let $x \in \mathcal{X} = [0, 1]^D$ and $B(x, \epsilon) = \{x' \in \mathcal{X} : \|x' - x\|_\infty \leq \epsilon\}$. We decompose a network as $f = h \circ \Phi$, where $\Phi : \mathcal{X} \rightarrow \mathbb{R}^d$ is a continuous feature map and $h : \mathbb{R}^d \rightarrow \mathbb{R}^C$, $h(z) = Wz + b$, is the classifier for C classes. We use subscripts t and s for the frozen natural teacher and the student. Both networks share the same architecture, with fixed network statistics for the analysis; feature distances are Euclidean unless stated otherwise.

3.2 DO NATURAL TEACHERS NEED SOFTENED TARGETS?

Can the natural teacher’s concentrated clean predictions provide effective adversarial supervision if softened? Motivated by soft-label robust distillation (Goldblum et al., 2020; Zi et al., 2021; Wu et al., 2024; Zhao et al., 2024), we first vary target concentration before changing the representation being transferred. Let

$$q_t^\tau(x) = \text{softmax}(f_t(x)/\tau), \quad p_s^\tau(x') = \text{softmax}(f_s(x')/\tau). \quad (1)$$

We compare softening only the teacher target with conventional shared-temperature KD:

$$D_{\text{KL}}(q_t^\tau(x) \| p_s^1(x')), \quad \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')). \quad (2)$$

As $\tau \rightarrow \infty$, the teacher target approaches a uniform distribution in both cases, but the scaled shared-temperature loss retains logit information. For fixed logits and $\bar{f} = f - C^{-1} \mathbf{1} \mathbf{1}^\top f$,

$$\lim_{\tau \rightarrow \infty} \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')) = \frac{1}{2C} \|\bar{f}_t(x) - \bar{f}_s(x')\|_2^2. \quad (3)$$

This limit motivates direct regression onto clean teacher logits; we also examine regression onto the full clean feature, which defines

$$\mathcal{L}_{\text{anchor}}(x) = \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2. \quad (4)$$

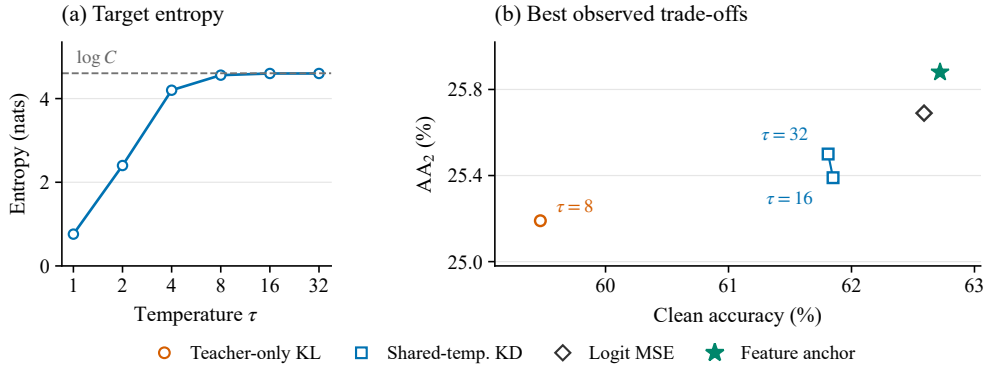


Figure 2: Target comparison on CIFAR-100 (ResNet-18). (a) Teacher-target entropy approaches the uniform value as τ increases. (b) Nondominated settings within each KD variant, alongside fixed-head logit and feature regression. The full temperature sweep is in Table 11.

We train for 50 epochs at $\epsilon = 8/255$. Across the full sweep, teacher-only softening improves robustness until the target becomes nearly uniform, but changes clean accuracy little. Shared-temperature KD avoids this turnover and approaches logit regression (Figure 2). Feature regression is only 0.13 clean and 0.19 AA points higher: both regression targets are effective without temperature selection. Moreover, KD updates the classifier whereas regression uses a fixed head, so this comparison evaluates training designs rather than isolating the target alone.

The comparison points to a fixed clean target rather than a particular temperature. We use the full feature to preserve the representation supplied to the inherited classifier, including directions that logit regression need not constrain. The small gap above does not establish a feature-specific performance advantage; it motivates examining what the single anchor controls.

3.3 WHAT THE ANCHOR CONTROLS

The fixed target links agreement with the teacher at the clean input to variation of the student’s features around it. To make this relation precise, let $x \in \mathcal{X}$ and $B = B(x, \epsilon)$, and define

$$L = \max_{x' \in B} \|\Phi_s(x') - \Phi_t(x)\|_2, \quad F = \|\Phi_s(x) - \Phi_t(x)\|_2, \quad O = \max_{x', x'' \in B} \|\Phi_s(x') - \Phi_s(x'')\|_2. \quad (5)$$

Proposition 1 (Fixed-anchor decomposition). *For every $x \in \mathcal{X}$ and $\epsilon \geq 0$,*

$$L \leq F + O \leq 3L. \quad (6)$$

Proof. For any $x' \in B$, the triangle inequality gives $\|\Phi_s(x') - \Phi_t(x)\|_2 \leq \|\Phi_s(x') - \Phi_s(x)\|_2 + F \leq O + F$; maximizing over x' yields $L \leq F + O$. Moreover, $x \in B$ implies $F \leq L$, and, for any $x', x'' \in B$, applying the triangle inequality through $\Phi_t(x)$ gives $\|\Phi_s(x') - \Phi_s(x'')\|_2 \leq 2L$. Hence $O \leq 2L$ and $F + O \leq 3L$. $\square$

If $L \leq \delta$, then the clean student feature is within δ of the teacher and the student’s feature diameter over the ball is at most 2δ , without assuming that the teacher itself is stable. The inherited linear head also bounds logit deviation by $\|W_t\|_{2 \rightarrow 2} L$, although classification depends on the teacher’s clean margin. The same argument applies to norm-based logit regression: it explains joint control by a fixed target, not a feature-specific advantage. Finite-step PGD does not certify L .

3.4 HOW DOES THE TEACHER CHECKPOINT AFFECT TRANSFER?

The bound holds for any fixed target, but cannot say which teacher will produce the best student. The most obvious choice is the checkpoint with the highest clean accuracy. We test this by training the same student procedure from different checkpoints of a natural teacher. Each checkpoint provides the initialization, feature target, and classifier, so the comparison measures their combined effect.

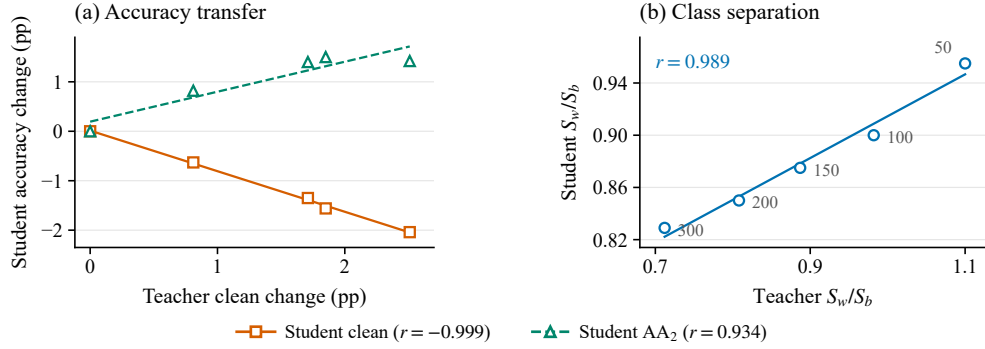


Figure 3: Teacher checkpoint sweep on CIFAR-100 (ResNet-18). (a) Accuracy changes relative to the 50-epoch teacher. (b) Within-to-between scatter ratios S_w/S_b ; lower means stronger class separation, and labels denote teacher epochs. Lines are least-squares fits. Exact values are in Table 12.

From 50 to 300 teacher epochs, teacher clean accuracy gains 2.51 points, whereas student clean accuracy falls by 2.04 and student AA rises from 24.38% to 25.80% (Figure 3a). The later checkpoint does not improve both student accuracies: it moves the operating point toward robustness. Nor is the shift strictly monotone; the 200-epoch student has slightly higher AA than the 300-epoch student.

An independently trained teacher trajectory shows the same pattern: from 40 to 300 epochs, teacher clean accuracy rises from 75.57% to 78.27%, while student clean accuracy falls from 64.36% to 62.34% and AA rises from 23.86% to 25.94% (Table 1). Both trajectories therefore reproduce a change in the student’s trade-off, rather than a rule that a later checkpoint is always better.

Table 1: Independent CIFAR-100 teacher trajectory (ResNet-18; 50-epoch students). Full results are in Table 24.

Teacher epoch	Teacher clean	Student clean	Student AA	Student NRR
40	75.57	64.36	23.86	34.81
100	77.27	63.68	25.61	36.53
200	77.90	63.02	26.16	36.97
300	78.27	62.34	25.94	36.64

To examine what accompanies this shift, we measure the within-to-between scatter ratio S_w/S_b on unit-normalized clean features. Teacher and student ratios decrease together (Figure 3b), consistent with stronger class separation. This association does not identify which change in the teacher causes the student shift.

The checkpoint sweep entangles training stage and teacher accuracy. To see whether accuracy alone describes transfer, we also compare a label-smoothed and a mixup teacher with nearly equal clean accuracy. Their students differ by 4.18 clean points but only 0.02 AA points (Table 2). Thus even at a similar teacher accuracy, the inherited representation can lead to a different student. The teachers’ feature sensitivities differ in the same order as student clean accuracy, but one matched pair cannot establish a general predictor.

Table 2: Accuracy-matched teachers and student transfer on CIFAR-100 (ResNet-18, 50-epoch students). Feature sensitivity is measured on held-out data.

	Teacher		Student	
	Clean	Sensitivity	Clean	AA
Natural training				
Label smoothing 0.1	78.76	3.16	59.62	25.72
Mixup	78.62	4.82	63.80	25.70

These results make teacher accuracy an incomplete guide to student performance, not a checkpoint selection rule. We use the 200-epoch teacher for subsequent experiments and turn to a choice that

remains even when the teacher is fixed: the same pixel radius need not change the anchor loss equally across examples.

4 METHOD

CFA combines a clean-feature training objective with sample-dependent perturbation budgets. The chosen teacher fixes the reference representation; radius allocation determines the neighborhood over which each example is trained to preserve it.

4.1 CLEAN FEATURE ANCHORING

Let Φ_t and h_t be the frozen feature map and classifier of the natural teacher. The student inherits both, $\Phi_s \leftarrow \Phi_t$ and $h_s = h_t$, after which only Φ_s is optimized. For a clean example x , CFA solves

$$\begin{aligned} x_{\text{adv}} &= \arg \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2, \\ \mathcal{L}_{\text{CFA}} &= \mathbb{E}_x [\|\Phi_s(x_{\text{adv}}) - \Phi_t(x)\|_2^2]. \end{aligned} \quad (7)$$

We approximate the inner maximization with randomly initialized PGD against the fixed target $\Phi_t(x)$ and minimize the same discrepancy over the backbone. This adversarial-training stage uses no label loss; inference uses $h_t \circ \Phi_s$. Algorithm 1 gives the complete procedure.

4.2 SENSITIVITY-MATCHED ϵ

A uniform pixel radius gives every example the same input-space budget, but not the same potential increase in anchor loss. We therefore allocate radii using the local sensitivity of the objective used for both attack generation and student training. The batch’s mean radius remains fixed; only its assignment across examples changes. For $\ell_{\text{anc}}(x'; x_i) = \|\Phi_s(x') - \Phi_t(x_i)\|_2^2$, define

$$g_i = \|\nabla_{x'} \ell_{\text{anc}}(x'; x_i)\|_1|_{x'=x_i}, \quad \max_{\|\delta\|_\infty \leq \epsilon} \langle \nabla_{x'} \ell_{\text{anc}}, \delta \rangle = \epsilon g_i. \quad (8)$$

The linearized worst-case loss increase under radius ϵ_i is therefore $\epsilon_i g_i$. To keep this local increase comparable across examples, a more sensitive example should receive a smaller radius. For $g_i > 0$, equalizing the products subject to $\sum_i \epsilon_i = N\epsilon$ gives

$$\epsilon_i = N\epsilon \frac{g_i^{-1}}{\sum_j g_j^{-1}}, \quad \text{or more generally} \quad \epsilon_i \propto g_i^{-p}, \quad (9)$$

where $p = 0$ recovers a uniform radius and $p = 1$ gives inverse-sensitivity allocation. The conserved sum separates redistribution from a change in the mean training budget. Unlike a sample-weighted outer loss, this rule changes the perturbations seen during training without changing the coefficient of each example in the training average. It is a local approximation to loss growth, not a claim that nonlinear adversarial losses become equal.

We implement $p = 1$ with an ℓ_2 gradient norm, measured by one backward pass at the clean input. The ℓ_1 norm follows exactly from the linearized ℓ_∞ problem, but the ℓ_2 proxy performs slightly better here (Appendix E). Thus the first-order argument motivates the rule without uniquely determining its implementation. Clipping and renormalization preserve the batch budget, and PGD step sizes scale with the assigned radii; implementation details are in Appendix E. The key test is whether sensitivity assigns the budget usefully, not merely whether varying radii helps. Accordingly, Table 8 compares uniform and alternative signals, including a control that keeps the exact multiset of sensitivity-derived multipliers but assigns them by sample difficulty. If the spread of radii alone explains the gain, this reassignment should preserve it; otherwise, which example receives each radius matters.

5 EXPERIMENTS

We test CFA across datasets and architectures, then examine the clean-feature objective and the allocation of perturbation radii. We report the contribution of weight averaging and AWP separately.

5.1 EXPERIMENTAL SETTINGS

We use CIFAR-10, CIFAR-100 (Krizhevsky et al., 2009), and Tiny-ImageNet-200 (Le & Yang, 2015). The natural teacher is trained on clean data for 200 epochs and then frozen (Table 22); its clean accuracies are 95.32%, 77.66%, and 66.29%, respectively, and its AA is zero on all three datasets. Unless stated otherwise, the ResNet-18 student is initialized from that teacher and trained with AdamW, a one-cycle schedule, and 10-step PGD, with sensitivity matching ($p = 1$), weight averaging, and AWP. The main CIFAR results use 150 epochs at the standard $\epsilon_{\text{train}} = 8/255$; ablations use 100 epochs and, where stated, $\epsilon_{\text{train}} = 8.8/255$. The Tiny-ImageNet and WideResNet runs also train at 8.8/255, which Table 17 shows moves the operating point along the trade-off rather than improving it: at 8/255 CIFAR-100 gives 63.89/27.78 against 62.17/28.86 at 8.8/255, so a single radius cannot be read as tuning. Baselines run their authors’ recipes, except that the main ARREST row uses 150 epochs with our optimizer, WA, and AWP; its 20-epoch recipe appears in Table 6. Detailed settings are in Appendix F.

We report clean and AA accuracy, their mean (Avg), and harmonic mean (NRR) on the full test set at $\epsilon = 8/255$. Distillation baselines receive our natural teacher. Evaluation details and results obtained with other teachers are in Appendices F and I and table 29.

5.2 MAIN RESULTS

Table 3: ResNet-18 results at ℓ_∞ , $\epsilon = 8/255$. All values are measured in our framework. T denotes the teacher assumed by the method: none (–), a natural model (nat.), or the method’s own weight average (self).

Method	T.	CIFAR-10				CIFAR-100			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	84.54	43.57	64.06	57.50	57.36	20.34	38.85	30.03
TRADES	–	81.71	49.16	65.44	61.39	55.28	23.55	39.42	33.03
MART	–	80.41	45.56	62.99	58.16	53.86	22.72	38.29	31.96
ARD	nat.	84.58	43.75	64.16	57.67	57.61	20.24	38.93	29.96
RSLAD	nat.	85.15	42.91	64.03	57.06	59.68	21.30	40.49	31.40
AdaAD	nat.	85.39	46.22	65.81	59.98	59.79	23.19	41.49	33.42
AdaAD + IGDM	nat.	85.01	45.50	65.26	59.27	48.25	19.48	33.87	27.75
LBGAT	nat.	–	–	–	–	56.46	26.02	41.24	35.62
ARREST	nat.	85.06	46.42	65.74	60.06	64.66	26.40	45.53	37.49
HAT	nat.	84.59	48.81	66.70	61.90	60.67	24.45	42.56	34.85
ADR	self	83.36	49.72	66.54	62.29	59.93	27.20	43.57	37.42
RPAT++	–	82.41	50.75	66.58	62.82	55.93	27.36	41.64	36.74
CFA (ours)	nat.	87.37	51.33	69.35	64.67	64.75	27.98	46.37	39.07

CFA attains the highest AA and NRR on both CIFAR datasets (Table 3). On CIFAR-100 it improves over ARREST by 1.58 AA points at comparable clean accuracy (64.75% versus 64.66%), and exceeds ADR, the next closest baseline in NRR, by 1.65 points. On CIFAR-10, its NRR exceeds the closest baseline by 1.85 points.

5.3 GENERALIZATION ACROSS DATA AND MODEL SCALE

On Tiny-ImageNet, CFA attains the highest Avg and NRR while maximizing neither endpoint: ARREST stays higher on clean accuracy and ADR is 0.31 points higher on AA (Table 4). The teacher-checkpoint effect transfers here too, since the 80-epoch teacher moves the student from 55.16/20.54 to 57.08/18.96 clean/AA. On CIFAR-10 with WideResNet-34-10 CFA is highest on both axes, matching ADR’s clean accuracy (88.67% versus 88.64%) with 2.95 points more AA. On CIFAR-100 with the same network, CFA reaches 65.04/30.14 clean/AA, against ADR’s 64.50/29.04 (Table 21).

Table 4: A larger dataset and a larger network, both measured in our framework: Tiny-ImageNet-200 with ResNet-18, and CIFAR-10 with WideResNet-34-10.

Method	T.	Tiny-ImageNet, ResNet-18				CIFAR-10, WRN-34-10			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	49.72	14.45	32.09	22.39	86.00	45.71	65.86	59.69
TRADES	–	46.98	15.95	31.46	23.83	85.13	48.99	67.06	62.19
MART	–	46.01	16.90	31.46	24.73	85.29	47.06	66.18	60.65
ARD	nat.	49.74	14.48	32.11	22.43	86.35	45.52	65.94	59.61
RSLAD	nat.	51.26	17.02	34.14	25.56	86.41	45.80	66.10	59.87
AdaAD	nat.	51.59	16.79	34.19	25.33	86.80	46.84	66.82	60.85
LBGAT	nat.	51.04	18.47	34.75	27.13	81.56	51.79	66.68	63.35
ARREST	nat.	59.38	16.06	37.72	25.28	87.54	48.83	68.19	62.69
HAT	nat.	52.57	17.28	34.93	26.01	87.15	53.21	70.18	66.07
ADR	self	47.66	20.85	34.25	29.01	88.64	52.34	70.49	65.82
CFA (ours)	nat.	55.16	20.54	37.85	29.93	88.67	55.29	71.98	68.11

5.4 WHAT PRODUCES THE GAIN?

Anchor and label supervision. Replacing the anchor by label cross-entropy under the same base regime reduces clean and AA by 6.69 and 5.91 points, and giving cross-entropy its own optimizer and schedule narrows the gap without closing it, leaving CFA ahead by 4.63 clean and 4.45 AA points (Table 5).

Table 5: Objective control on CIFAR-100 with ResNet-18, 100 epochs, no weight averaging or AWP, $p = 0$, and $\epsilon_{\text{train}} = 8.8/255$.

Objective	Clean	AA	NRR
Anchor	61.29	25.36	35.88
Label cross-entropy, same recipe	54.60	19.45	28.68
Label cross-entropy, own recipe	56.66	20.91	30.55

Initialization and schedule controls change NRR by at most 0.22 (Table 25). Even under the full stack, label cross-entropy remains below CFA in both clean and AA accuracy. Applying the same additions to the baselines’ own recipes helps most but does not close the gap (Table 6).

Regression and probability targets. Logit regression remains effective with WA and AWP: at a uniform 8.8/255 and 100 epochs, logit MSE reaches 60.36/28.02 clean/AA against feature regression’s 60.42/28.42 (Table 13), so with the base-regime comparison of Section 3.2, effective clean-target regression is not confined to feature space. Teacher-only KL reaches 56.97/26.96 at its best tested temperature under the full recipe (Table 28), a comparison that does not isolate the target from classifier and radius choices.

Table 6: Adding teacher initialization, weight averaging, and AWP to baseline recipes on CIFAR-100 (ResNet-18).

Method	Own recipe			+ teacher init + WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
PGD-AT	57.36	20.34	30.03	59.85	25.77	36.03
TRADES	55.28	23.55	33.03	56.16	24.56	34.17
MART	53.86	22.72	31.96	50.97	25.13	33.66
ARD	57.61	20.24	29.96	60.19	26.10	36.41
RSLAD	59.68	21.30	31.40	63.44	25.55	36.43
AdaAD	59.79	23.19	33.42	57.21	27.10	36.78
AdaAD + IGDM	48.25	19.48	27.75	54.24	25.56	34.75
Consistency-AT	57.53	20.01	29.69	60.74	25.38	35.80
LBGAT	56.46	26.02	35.62	53.91	25.16	34.31
ADR	59.93	27.20	37.42	59.84	27.84	38.00
HAT	60.67	24.45	34.85	57.63	24.74	34.62
ARREST	66.44	22.99	34.16	64.95	23.02	33.99
CFA (ours)	—	—	—	64.75	27.98	39.07

The additions raise NRR for nine of twelve baselines, yet none reaches CFA; ADR is closest at 38.00 versus 39.07. ARREST’s rows here use its 20-epoch recipe, unlike its 150-epoch comparison in Table 3. ADR already uses a SAM-style weight perturbation in its own recipe.

5.5 RADIUS ALLOCATION AND TRAINING COMPONENTS

Does allocation help in the final recipe? Holding the teacher, training schedule, mean radius, weight averaging, and AWP fixed, we replace the uniform radius with sensitivity matching in the 150-epoch CIFAR-100 run. Clean accuracy increases by 2.35 points, while AA changes by 0.14 points (Table 7). The gain therefore persists at the configuration reported in Table 3; it is not confined to the shorter ablation runs. The comparison primarily supports a clean-accuracy gain at similar robustness; the 0.14-point AA difference is too small to interpret without repeated runs of both arms.

Table 7: Radius allocation in the final CIFAR-100 recipe (ResNet-18, 150 epochs, $\epsilon_{\text{train}} = 8/255$). The two runs differ only in p .

Radius	Clean	AA	NRR
Uniform ($p = 0$)	62.40	27.84	38.50
Sensitivity-matched ($p = 1$)	64.75	27.98	39.07

Which signal should allocate the radius? At 100 epochs, we compare the sensitivity signal with alternative assignments of the same mean budget (Table 8). Reassigning the *same radii* by sample difficulty loses 0.80 AA points relative to sensitivity, so varying radii alone does not explain the gain. Difficulty and margin signals also have lower NRR. These are controls on the allocation signal, not reproductions of other adaptive-radius algorithms.

Table 8: Radius-allocation signals under the same mean radius, CIFAR-100, ResNet-18, 100 epochs with WA and AWP. $\text{sd}(w)$ measures multiplier dispersion.

Allocation	$\text{sd}(w)$	Clean	AA	NRR
Uniform	0	60.42	28.42	38.66
Sensitivity	0.32	62.17	28.86	39.42
Same multipliers, difficulty order	0.32	60.90	28.06	38.42
Difficulty magnitude	0.39	61.27	28.12	38.55
Logit margin	0.42	61.07	27.79	38.20

Weight averaging and AWP. We also apply weight averaging and AWP in the final training recipe. In the ordered ablation, they add 1.95 and 1.17 AA points, respectively (Table 26). This sequence does not isolate interactions; radius and stack comparisons are in Tables 17 and 18.

Attack and repeat-seed diagnostics are in Appendices F and I.

6 CONCLUSION

A natural teacher can guide adversarial training through a single clean-feature target, without a separate label loss. Yet the teacher checkpoint matters: later natural training shifts the student toward robustness in two trajectories, and similar teacher clean accuracy need not yield similar student clean accuracy. With the teacher fixed, sensitivity-based radii improve clean accuracy at nearly unchanged AutoAttack accuracy in the final recipe. CFA thus separates three decisions often treated together in natural-teacher distillation: the target, the teacher, and the perturbation budget.

A ALGORITHM

Algorithm 1 Clean Feature Anchoring (CFA)

```

1: Input: dataset  $\mathcal{D}$ , radius  $\epsilon$ , attack steps  $K$ , step size  $a$ , exponent  $p$ , clipping thresholds  $[w_{lo}, w_{hi}]$ 
2: Output: robust student  $h_t \circ \Phi_s$ 
3:
4: {Phase 1 — natural self-teacher. The only stage that uses labels.}
5: Train  $f_t = h_t \circ \Phi_t$  on  $\mathcal{D}$  with standard cross-entropy; freeze it
6:
7: {Phase 2 — anchored adversarial training. No labels, no head update.}
8:  $\Phi_s \leftarrow \Phi_t$  {student initialized at the teacher}
9: for epoch = 1 to  $T$  do
10:   for minibatch  $\{x_i\}_{i=1}^N \subset \mathcal{D}$  do
11:      $\phi_i \leftarrow \Phi_t(x_i)$  {teacher read once, at the clean input}
12:      $g_i \leftarrow \max(\|\nabla_x \|\Phi_s(x_i) - \phi_i\|_2, 10^{-12})$  {one backward pass}
13:      $v_i \leftarrow \text{clip}((\bar{g}/g_i)^p, w_{lo}, w_{hi})$ ;  $w_i \leftarrow N v_i / \sum_j v_j$ 
14:      $\epsilon_i \leftarrow \epsilon w_i$ ;  $a_i \leftarrow a w_i$  { $\sum_i \epsilon_i = N\epsilon$ }
15:      $x_i^0 \leftarrow x_i + \eta_i$ 
16:     for  $k = 0$  to  $K - 1$  do
17:        $x_i^{k+1} \leftarrow \Pi_{B(x_i, \epsilon_i)}(x_i^k + a_i \text{sign } \nabla_x \|\Phi_s(x_i^k) - \phi_i\|_2)$ 
18:     end for
19:      $\mathcal{L} \leftarrow \frac{1}{N} \sum_i \|\Phi_s(x_i^K) - \phi_i\|^2$ 
20:      $\Phi_s \leftarrow \Phi_s - \mu \nabla_{\Phi_s} \mathcal{L}$  { $h_t$  is never updated}
21:   end for
22: end for

```

B TWO PARTIAL RESULTS ON THE ROBUSTNESS SIDE

Natural-teacher guidance is shared with prior work; our question is whether a fixed clean target can support adversarial training without an additional label objective. In CFA, the teacher specifies the target while the student’s inner maximization enforces matching under perturbations. Teacher vulnerability alone therefore does not determine the student’s response: matching a clean target is different from reproducing the teacher at perturbed inputs.

The two results below examine this distinction in restricted settings. The first studies robust and non-robust features in a two-feature model; the second relates anchor error to a classification margin. They supplement the fixed-anchor decomposition, but do not predict the measured accuracy or establish superiority over other regression targets.

B.1 IN THE TWO-FEATURE MODEL, THE NON-ROBUST COEFFICIENT IS ZEROED RATHER THAN SHRUNK

The setting is the two-feature model of Tsipras et al. (2018), which is where the objection is usually posed. One feature x_1 agrees with the label with probability $p = 0.95$ and cannot be attacked. A further d features $z_j \sim \mathcal{N}(\eta y, 1)$ are individually almost uninformative, are jointly close to perfectly informative through their mean $m(z)$, and can all be moved by an ℓ_∞ adversary. A naturally trained network therefore reads $m(z)$: near-perfect clean accuracy, no robustness whatever. That is our teacher’s profile, so $\Phi_t = m(z)$.

Write the student’s feature as $\Phi_s = ax_1 + bm(z)$. Then b is exactly the quantity the objection is about: how much of the teacher’s fragile route the student takes over. If the objection is right, minimizing our objective should leave b large.

Proposition 2. *With the inner maximum exact, $J(a, b) = \mathbb{E}[R^2] + 2\epsilon|b| \mathbb{E}|R| + \epsilon^2 b^2$ with $R = ax_1 + (b - 1)m$. Its one-sided slopes at $b = 0$ differ by $4\epsilon \mathbb{E}|R(a, 0)|$; above a threshold ϵ_0 , every minimizer has $b^* = 0$.*

Proof. The adversary reaches Φ_s only through m , and only through the mean perturbation $\bar{\delta} = \frac{1}{d} \sum_j \delta_j$, which ranges over $[-\epsilon, \epsilon]$ as $\|\delta\|_\infty \leq \epsilon$. The inner objective is therefore $\max_{|\bar{\delta}| \leq \epsilon} (R + b\bar{\delta})^2 = (|R| + \epsilon|b|)^2$, since a scalar quadratic is maximized at an endpoint and the sign of $\bar{\delta}$ is free. Taking expectations and expanding the square gives the display.

Fix a and differentiate in b . The term $\mathbb{E}[R^2]$ and $\epsilon^2 b^2$ are smooth, while $2\epsilon|b| \mathbb{E}[R]$ contributes $\pm 2\epsilon \mathbb{E}[R]$ as $b \rightarrow 0^\pm$. The one-sided derivatives at $b = 0$ are thus $2\mathbb{E}[R(a, 0) m] \pm 2\epsilon \mathbb{E}[R(a, 0)]$, so $b = 0$ is a minimum along b exactly when

$$|\mathbb{E}[R(a, 0) m]| \leq \epsilon \mathbb{E}[R(a, 0)]. \quad (10)$$

At $b = 0$ the objective in a is $\mathbb{E}(ax_1 - m)^2$. Writing $q = 2p - 1$ for the mean label-alignment of x_1 , it is minimized at $a^* = \mathbb{E}[x_1 m] / \mathbb{E}[x_1^2] = q\eta$, and there $\mathbb{E}[R(a^*, 0) m] = q^2 \eta^2 - (\eta^2 + \frac{1}{d}) = -c$ with

$$c = \eta^2 v + \frac{1}{d}, \quad v = 1 - q^2. \quad (11)$$

Equation (10) therefore holds for every

$$\epsilon \geq \epsilon_0 = \frac{c}{\mathbb{E}[q\eta x_1 - m]}, \quad (12)$$

which is the stated threshold. For fixed x_1, m and $\bar{\delta} \in [-\epsilon, \epsilon]$, $(ax_1 + (b-1)m + b\bar{\delta})^2$ is convex in (a, b) . Its maximum over $\bar{\delta}$ and then its expectation are convex, so J is jointly convex. At $(a^*, 0)$ the a -derivative vanishes, and Equation (10) puts zero in the b -subgradient; hence this point is a global minimizer. If $\epsilon > \epsilon_0$, the directional derivative from $(a^*, 0)$ is strictly positive in every direction with nonzero b . Along $b = 0$, $\mathbb{E}(ax_1 - m)^2$ has a unique minimizer. Thus every global minimizer has $b = 0$ above the threshold. $\square$

Why the coefficient is deleted rather than shrunk. The adversary contributes $2\epsilon|b| \mathbb{E}[R]$ to the loss — linear in $|b|$, not quadratic in b . A quadratic penalty pushes a coefficient toward zero and never arrives; a penalty on $|b|$ is kinked at the origin and sets the coefficient to zero outright once its slope dominates, which is the same distinction that makes the lasso sparse and the ridge not. What produces the kink here is that the student is asked to reproduce a *value* rather than a *route*: the fragile features carry that value on clean data but not inside the ball, so past a certain radius carrying any of them costs more than it pays. The teacher supplies what to match, and the inner maximization discards how the teacher got there.

At $d = 512$ the model's constants give $c = 7.89 \times 10^{-3}$ and $\mathbb{E}[q\eta x_1 - m] = 0.0529$, hence $\epsilon_0 = 0.843 \eta$ — inside the bracket $(0.5\eta, 1.0\eta)$ that the Monte-Carlo minimization returns, and far below the 2η at which the model is usually stated.

The model assumes x_1 is unperturbable and z fully perturbable, so some suppression of z is built in. The proposition establishes the exact threshold behavior within this restricted model; it does not describe the geometry of real image classifiers.

That the exactness is the kink's doing, and not a general consequence of adding an adversary, is visible from dropping it. With the $|b|$ term removed, $J_{\text{quad}} = \mathbb{E}[R^2] + \epsilon^2 b^2$; minimizing out a and writing $u = 1 - b$ collapses it to $u^2 c + \epsilon^2(1 - u)^2$, so

$$b_{\text{quad}}^* = \frac{c}{c + \epsilon^2}, \quad (13)$$

which is strictly positive at every radius. Against the exact minimizer ($d = 512$, Monte-Carlo, $n = 2 \times 10^5$):

ϵ	b^* (exact)	b_{quad}^* (kink dropped)
0.5η	0.510	0.502
1.0η	0.000	0.202
2.0η	0.000	0.060
4.0η	0.000	0.016

The two columns agree below ϵ_0 and separate above it: the quadratic surrogate shrinks the non-robust coefficient and never deletes it, which is the ordinary picture, while the kinked objective sets it to zero and keeps it there.

Since every ϵ in J multiplies b , at $b^* = 0$ the objective reduces to $\mathbb{E}(ax_1 - m)^2$ and no longer depends on the radius. This saturation is a property of the two-feature model, not a prediction of the empirical radius sweep in Table 17.

The two-feature model cannot explain the measured clean-robust trade-off: it has only a robust and a fragile direction. It shows the narrower point that a robust student need not copy the fragile route by which its natural teacher computes the clean target.

B.2 WHAT ROBUST ACCURACY TRACKS IS THE LOSS MEASURED AGAINST A MARGIN

Writing $\gamma_t(x)$ for the teacher’s margin at x and D_y for the relevant decision-boundary constant, Cauchy–Schwarz gives that $\gamma_t(x) > D_y \cdot L(x)$ implies the student is robustly correct at x . The certificate is vacuous at the scales we train at, but the ratio DL/γ_t tracks both axes across the teacher ladder of Figure 3. What it does not do is predict the size of the effect, and it saturates before AutoAttack does.

C SUPPLEMENTARY ANALYSIS OF THE CLEAN-FEATURE ANCHOR

C.1 TEACHER EVALUATION ON PERTURBED INPUTS

The comparison in Section 3.2 evaluates the teacher at clean inputs. Here we examine why its responses to perturbed inputs may be unsuitable as supervisory targets. Under $\epsilon = 8/255$, the natural teacher’s feature rotates by 63.8° and its norm grows by a factor of 2.45. Its representation can therefore change substantially within the perturbation ball.

As a control, replacing the clean target $\Phi_t(x)$ with $\Phi_t(x')$ makes an exact copy of the teacher, $\Phi_s = \Phi_t$, a zero-loss solution even when the teacher is sensitive to perturbations. This variant yields 0.00% AutoAttack accuracy (Table 9). With a fixed clean target, copying the teacher instead incurs a loss wherever its features change within the ball.

Table 9: Where the teacher is evaluated. CIFAR-100, ResNet-18, 50 epochs; the two rows differ only in the teacher input used by the loss and inner maximization.

Teacher target	Clean	AA	NRR
$\Phi_t(x)$	61.33	26.19	36.71
$\Phi_t(x_{\text{adv}})$	76.11	0.00	0.00

IGDM (Lee et al., 2025) uses the finite difference $f_t(x + \delta) - f_t(x - \delta)$ to approximate the teacher’s response along the perturbation direction. Its accuracy depends on local linearity. The reported Taylor remainder proportion is 0.012 for adversarially trained teachers; we obtain 0.0111 for an adversarially trained ResNet-18 and 0.4763 for the natural teacher (Table 14). For the natural teacher, the finite difference has norm 14.14, compared with $\|f_t(x)\| = 18.36$, and its softmax assigns 0.917 probability to one class. These measurements indicate a less accurate local linear approximation and a highly concentrated target distribution.

We also evaluate published distillation objectives with the natural teacher replacing their robust teacher, using each method’s published recipe (Table 19). All evaluated objectives perform below adversarial training initialized from that teacher. This comparison complements the target sweep in Section 3.2, although differences among training recipes prevent it from isolating the target alone.

C.2 ADDITIONAL EVIDENCE FOR THE ANCHOR OBJECTIVE

The geometry and displacement diagnostics below use the earlier 100-epoch `ladder_p0_100ep` anchor and `abl_ce_nostack` checkpoints. The anchor’s backbone uses raw feature regression, while a detached KD term trains its classifier; this is distinct from the frozen-head models in Table 5. Both diagnostics use true-label PGD with 10 steps at radius $8.8/255$. The displacement script samples training batches, whereas the class-geometry script samples test batches. Consequently, they are complementary descriptive measurements rather than a joint causal test on the final model.

Proposition 1 bounds variation within an anchored model, but does not predict a smaller variation than label training. Under the matched true-label PGD protocol, the anchor and label-trained students move by 0.275 and 0.254 of their clean feature norms. These displacements are not the worst-case diameter O and provide no evidence of smaller relative displacement under anchoring. Instead, the stronger difference is in class organization: the anchor has a $5.5\times$ smaller clean S_w/S_b and a $2.7\times$ smaller ratio under attack (Table 10).

Table 10: Feature geometry under the same true-label PGD attack, using the earlier base-regime models described above. Angles are measured on unit-normalized features. *Own* is the sample-to-own-centroid angle; *gap* is the mean angle from each class centroid to its nearest other-class centroid.

	Clean			Adversarial		
	S_w/S_b	Own	Gap	S_w/S_b	Own	Gap
Anchor	0.898	28.1°	28.6°	2.681	33.3°	16.2°
Label CE	4.960	45.4°	21.6°	7.143	47.0°	18.3°

The classification benefit is evaluated in Section 5.4. Without weight averaging, AWP, or the sensitivity-matched radius, the anchor improves clean and AutoAttack accuracy over label cross-entropy by 6.69 and 5.91 points under the same training conditions. The gains are 4.63 and 4.45 points when cross-entropy uses its own optimizer and schedule (Table 5, Table 25). At the full training configuration, the AutoAttack gain over cross-entropy with weight averaging and AWP is 2.40 points. Under a matched attack, the students exhibit similar relative feature displacement, while their class separation differs by a factor of 5.5. This suggests a contribution from the learned geometry, but does not establish class separation as the sole cause of the robustness gain.

The checkpoint ladder in Figure 3 further separates within-class concentration from between-class separation on unit-normalized features. From the 50-epoch to the 300-epoch teacher, the mean angle from a sample to its class centroid decreases from 34.6° to 30.9°, whereas the nearest-other-centroid angle between class means increases from 35.7° to 44.9°. The corresponding students reproduce the latter change—their nearest-centroid separation increases from 25.4° to 29.8°—while their mean within-class angle remains nearly unchanged (27.9° to 28.2°). These measurements motivate interpreting the ladder through between-class geometry, but they remain observational because the teacher’s accuracy and geometry change together. The numerical values underlying the two analysis figures are collected below.

The target comparison uses the 50-epoch base regime at $\epsilon = 8/255$, without weight averaging or AWP.

Table 11: Numerical values for the target comparison in Figure 2. $H(q_t^\tau)$ is measured in nats, and NRR is the harmonic mean of clean and AA accuracy.

Target	τ	$H(q_t^\tau)$	Clean	AA	NRR
<i>Teacher-only temperature</i>					
KL	1	0.76	58.26	20.84	30.70
KL	2	2.40	59.33	22.79	32.93
KL	4	4.20	59.39	24.48	34.67
KL	8	4.56	59.47	25.19	35.39
KL	16	4.60	57.78	24.00	33.91
<i>Shared-temperature KD</i>					
KL	4	4.20	60.83	25.05	35.49
KL	8	4.56	60.99	25.25	35.71
KL	16	4.60	61.85	25.39	36.00
KL	32	4.60	61.81	25.50	36.11
Logit regression, fixed head	—	—	62.59	25.69	36.43
Feature anchor, fixed head	—	—	62.72	25.88	36.64

Table 12: Numerical values for the checkpoint trends in Figure 3.

Checkpoint	Teacher		Student		
	Clean	S_w/S_b	Clean	S_w/S_b	AA
50 epochs	75.81	1.100	64.28	0.955	24.38
100 epochs	76.62	0.982	63.65	0.900	25.20
150 epochs	77.52	0.887	62.93	0.875	25.78
200 epochs	77.66	0.808	62.72	0.850	25.88
300 epochs	78.32	0.712	62.24	0.829	25.80

C.3 SCOPE OF THE ANCHOR BOUND

Proposition 1 concerns exact maxima over the perturbation set. A loss evaluated using a finite attack does not certify these maxima. The factor 3 combines $F \leq L$ with $O \leq 2L$; the latter bound is attained when two student features lie on opposite sides of the target, each at distance L .

The argument applies to any fixed target in a normed space, including logits when matched using a norm distance. It explains how one anchor controls two quantities, but does not establish that features are superior to logits. The feature choice is motivated by Section 3.2. Likewise, the bound does not directly lower-bound robust classification accuracy: this requires information about the classifier and class margins, as discussed in Appendix B.

Joint control also does not imply that fidelity and stability can both be made arbitrarily small over the dataset. When perturbation balls overlap but their teacher targets differ, they impose conflicting constraints on the same student feature. The proposition describes how the objective controls the two errors without eliminating this possible conflict.

D ADDITIONAL RESULTS

The following tables provide additional results supporting the main experiments and supplementary analysis.

Regression under weight averaging and AWP. Table 13 compares logit and feature regression with the same natural teacher, inherited frozen classifier, AdamW optimizer, one-cycle schedule, WA, and AWP. Both use a uniform radius: neither run includes sensitivity-based allocation. Each objective is maximized to generate training perturbations and minimized to update the backbone, so the comparison includes its effect on both stages. Logit regression remains competitive with the training stack; the single-seed difference does not establish a general advantage for feature regression.

Table 13: Fixed-head regression with WA and AWP: CIFAR-100, ResNet-18, 100 epochs, uniform $\epsilon_{\text{train}} = 8.8/255$. Evaluation uses AA at $8/255$. Both losses sum squared errors across output coordinates and average over the batch.

Target	Clean	AA
Clean logits	60.36	28.02
Clean features	60.42	28.42

Table 14: Local linearity, which IGDM’s construction requires of the teacher: remainder proportion of the first-order Taylor expansion over an $\epsilon = 8/255$ ball, computed as in IGDM’s Sec. 3.1.

Network	Remainder proportion
IGDM’s robust teachers (LTD / BDM-AT / IKL-AT, published)	0.012 / 0.012 / 0.016
PGD-AT ResNet-18 (ours)	0.0111
Natural teacher (ours)	0.4763
Anchored student, $p = 0$	0.0029
Anchored student, shipped recipe	0.0202

Table 15: What to do with the classifier. CIFAR-100 base regime (50 epochs, raw features, no weight averaging, no AWP, $\epsilon = 8/255$), so that no stack component can absorb the difference.

	Clean	PGD-20	CW	AA	NRR
Logit target only, $\tau = 1$	58.26	22.62	22.79	20.84	30.70
Logit target only, $\tau = 4$	59.39	30.30	26.84	24.48	34.67
Logit target only, $\tau = 16$	57.78	31.57	26.11	24.00	33.91
Feature anchor + head KD, $\tau = 1$	62.34	26.34	26.11	23.93	34.58
Feature anchor + head KD, $\tau = 16$	62.61	32.47	27.32	25.61	36.35
Feature anchor, head untouched	62.72	28.69	27.80	25.88	36.64
head refit, adversarial CE	62.77	29.93	27.66	25.65	36.42
head refit, adv. CE + smoothing 0.1	62.57	30.43	27.55	25.66	36.39
head refit, clean CE	62.70	28.90	27.23	25.15	35.90

Normalization check. Because Section 3.4 analyzes angular geometry, we check whether discarding feature magnitude changes the training conclusion. Normalizing both sides changes NRR by 0.18 with weight averaging and AWP, while raw regression is better in the base regime. We therefore retain raw squared ℓ_2 as the default and use normalized features only for analysis; CFA does not rely on a cosine objective.

Table 16: Effect of feature normalization in the anchor loss. Normalizing both representations to the unit sphere removes their magnitudes and keeps only the angle, for which the squared Euclidean loss satisfies $\|\hat{\Phi}_s - \hat{\Phi}_t\|_2^2 = 2 - 2 \cos \angle(\Phi_s, \Phi_t)$ and is therefore equivalent to cosine loss up to an affine transformation. CIFAR-100, ResNet-18, 100 epochs; within each panel, the runs differ only in how the student and teacher features are normalized. The asymmetric row normalizes the student alone, placing the two sides at different scales, and is included only as a diagnostic.

	Clean	AA	NRR
<i>(a) With the stack: WA, AWP, $\epsilon_{\text{train}} = 8.8/255$</i>			
Raw features (squared ℓ_2)	60.21	28.14	38.35
Both normalized (cosine)	61.41	28.07	38.53
<i>(b) Base regime: no WA, no AWP, $\epsilon_{\text{train}} = 8/255$</i>			
Raw features (squared ℓ_2)	62.40	24.34	35.02
Both normalized (cosine)	61.52	22.90	33.38
Asymmetric: student normalized, teacher raw	61.28	22.98	33.43

Table 17: ϵ_{train} swept with everything else fixed. Evaluation is at $\epsilon = 8/255$ in every cell.

	Clean	PGD-20	CW	AA	NRR
<i>CIFAR-100</i>					
6/255	68.07	28.40	28.16	25.82	37.44
7/255	65.61	30.34	29.64	27.44	38.70
8/255	63.89	31.41	29.83	27.78	38.72
8.8/255 [†]	62.17	32.37	30.93	28.86	39.42
10/255 [†]	59.99	32.82	30.53	28.77	38.89
<i>CIFAR-10</i>					
6/255	89.64	49.29	50.20	47.88	62.42
7/255	88.30	51.43	52.64	50.27	64.07
8/255 [†]	87.22	52.43	53.55	51.15	64.48
8.8/255 [†]	84.96	52.98	53.94	51.74	64.31
10/255 [†]	83.29	53.25	53.88	51.79	63.87

Table 18: Effect of AWP on top of weight averaging across datasets and training radii. Every run uses the sensitivity-matched radius for 100 epochs; the AWP proxy uses $\gamma = 0.005$ after a 10-epoch warmup. Within each row, AWP is the only added component.

	WA			WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
<i>CIFAR-100</i>						
8/255	63.54	26.75	37.65	63.89	27.78	38.72
8.8/255	62.30	27.69	38.34	62.17	28.86	39.42
<i>CIFAR-10</i>						
8/255	86.67	49.20	62.77	87.22	51.15	64.48
8.8/255	85.16	50.15	63.13	84.96	51.74	64.31

Table 19: Published adversarial-distillation objectives supplied with our natural teacher. The first row of each method uses its published recipe; indented rows add our teacher initialization and then without weight averaging or AWP.

	CIFAR-100			CIFAR-10		
	Clean	AA	NRR	Clean	AA	NRR
ARD	57.61	20.24	29.96	84.58	43.75	57.67
+ teacher initialization	57.96	20.22	29.98	—	—	—
RSLAD	59.68	21.30	31.40	85.15	42.91	57.06
+ teacher initialization	59.87	20.71	30.77	—	—	—
AdaAD	59.79	23.19	33.42	85.39	46.22	59.98
+ teacher initialization	59.75	22.78	32.98	—	—	—
AdaAD + IGDM	48.25	19.48	27.75	85.01	45.50	59.27
+ teacher initialization	49.67	20.15	28.67	—	—	—
PGD-AT, teacher initialization	57.73	26.46	36.29	82.23	50.72	62.74
CFA, no WA/AWP	61.29	25.36	35.88	—	—	—

Table 20: RPAT++ given the rest of our stack. Its own recipe already carries weight averaging (EMA 0.999 from epoch 105 of 200), so the missing component is the AWP proxy, added at $\gamma = 0.005$ with warmup at 10% of the run and evaluated on its own loss; the last row also starts the run at our natural teacher. CIFAR-100, ResNet-18, our reproduction, reported at both the final and the PGD-20-selected weight-averaged checkpoint.

	Final			Best by PGD-20		
	Clean	AA	NRR	Clean	AA	NRR
RPAT++ (its own recipe)	57.05	26.61	36.30	55.93	27.37	36.74
+ AWP	57.48	28.05	37.70	57.50	27.92	37.60
+ AWP + teacher initialization	57.83	27.47	37.25	57.80	27.59	37.36

E THE ALLOCATION RULE AND ITS BUDGET SOLVE

E.1 WHERE THE RULE COMES FROM

The objective is optimized over a **pixel** ball, while what it measures is a displacement in feature space, and a fixed pixel radius moves that displacement by very different amounts across samples. The allocation follows from one elementary fact: for a differentiable loss the first-order movement available inside an ℓ_∞ ball of radius e is

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_x \mathcal{L}, \delta \rangle = e \|\nabla_x \mathcal{L}\|_1, \quad (14)$$

the dual norm of ℓ_∞ being ℓ_1 . Equalizing that movement across a batch subject to a fixed total budget $\sum_i \epsilon_i = N\epsilon$ gives $\epsilon_i g_i = \text{const}$, that is $\epsilon_i \propto 1/g_i$ with $g_i = \|\nabla_x \mathcal{L}(x_i)\|_1$. The same allocation is the max-min one: if two samples move by different amounts, shifting budget from the sample that moves more to the sample that moves less raises the minimum, so no unequal assignment is optimal. Taking $\epsilon_i \propto (\bar{g}/g_i)^p$ makes $p = 1$ that allocation, $p = 0$ the uniform-pixel one, and $p \in (0, 1)$ an interpolation.

Two approximations are worth declaring rather than leaving to be found. First, g_i is measured once, by a single backward pass at the *clean* x_i before the attack starts, so it is the sensitivity at the starting point and not along the PGD trajectory. Second, everything above is first order in ϵ . Neither is repaired anywhere in the method; the rule is a budget heuristic with a derivation, not an exactly solved subproblem.

We also do not claim the rule is specific to a directional objective. $\|\nabla_x \mathcal{L}\|_1$ is defined for any loss and the implementation reads whichever loss is configured; on the shipped raw- ℓ_2 anchor it equalizes a feature displacement rather than an angle, which is why the honest name is *sensitivity-matched* ϵ rather than an angular budget. What separates it from IAAT, MMA and CAT is not directionality but the signal: they allocate from a property of the sample, this allocates from the geometry the training loss is written in.

The separation is not one of dispersion, which is worth stating because a rule that simply spread the budget more aggressively would be the simpler explanation of any gain. Measured over training on the shipped configuration, the coefficient of variation of our sensitivity signal has median 0.56, against 0.75 for the difficulty score IAAT and CAT allocate from and 1.33 for the logit margin of MMA. The box of this section compresses all three before they reach the attack, leaving per-sample multipliers of spread 0.32, 0.39 and 0.42 respectively, and the ordering survives the compression. Ours is the least dispersed of the three signals either way, so the comparison in Table 8 is a comparison of which geometry the budget follows.

One discrepancy is ours to declare rather than the reader’s to find. Equation (14) gives the ℓ_1 norm and the implementation reads ℓ_2 , for no better reason than that the cell which fixed the recipe used it and every logged run since had to stay reproducible against that choice. The two are not the same allocation. Measured over training on the shipped configuration, the ℓ_1 signal’s coefficient of variation has median 0.68 against the ℓ_2 signal’s 0.56, and the per-sample multiplier the attack receives has spread 0.34 against 0.32 — so the norms really do distribute the budget differently.

They nonetheless land close together. Training the shipped recipe with `featdir_angepts_gnorm` set to 1 and nothing else changed gives 62.16/31.97/30.56/28.57 (NRR 39.15) against 62.17/32.37/30.93/28.86 (NRR 39.42): clean accuracy is identical to a hundredth and AutoAttack differs by 0.29, in favour of the norm we ship.

We do not call that a tie. It is twice the seed-to-seed spread of Appendix F, and it is the same order of magnitude as what the allocation buys in the first place: against the uniform radius, $p = 1$ gains +1.75 clean and +0.44 AutoAttack at ℓ_2 and +1.74 and +0.15 at ℓ_1 . So the rule’s benefit survives the norm in direction and almost entirely on the clean axis, while most of its AutoAttack half does not. The honest statement is that the norm is a real choice which we made before we could compare, that the comparison favours what we ship, and that a reader preferring the derived ℓ_1 would still get the clean gain and a third of the robust one.

The permuted row of Table 8 deserves its construction spelled out, since it is what makes the comparison a comparison of signals rather than of budgets. It takes the multiset of weights the gradient rule produces — the same values, the same clip $[0.5, 1.5]$, the same restored mean — and only permutes which example receives which, so that the larger radii go to the easier examples, which is the direction IAAT and CAT assign them. Reordering costs 0.80 AutoAttack against the rule it reorders and lands below even the uniform row on AutoAttack, CW and NRR alike.

E.2 CLIPPING AND RESTORING THE MEAN RADIUS

The reported configuration floors the gradient norm at 10^{-12} and computes $v_i = \text{clip}((\bar{g}/g_i)^p, 0.5, 1.5)$, followed by $w_i = Nv_i / \sum_j v_j$ and $\epsilon_i = \epsilon w_i$. PGD uses step size $a_i = \epsilon w_i$. This preserves the batch mean radius exactly. The clipping thresholds apply before normalization, so they are not hard bounds on the final multipliers. When all gradients vanish, the floor makes all g_i equal and the allocation uniform.

Optional exact-box variant. The implementation also supports an alternative that constrains the final multipliers. It is not enabled in the reported default configuration.

With $r_i = (\bar{g}/g_i)^p$ and box $[w_{\text{lo}}, w_{\text{hi}}] = [0.5, 1.5]$, the radii that satisfy both the box and the budget exactly are $\epsilon_i = \epsilon \cdot \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}})$ with t the solution of

$$\sum_{i=1}^N \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}}) = N. \quad (15)$$

The left side is continuous and nondecreasing in t from Nw_{lo} to Nw_{hi} , so a root exists and bisection finds it; when no clip is active it reduces to $t = N/\sum_i r_i$, that is, to restoring the mean. Clipping first and rescaling afterwards, the obvious alternative, does not preserve the box—the rescaling moves the clipped entries back out of it.

Table 21: WideResNet-34-10 on CIFAR-100 at $\epsilon = 8/255$, measured in our framework.

Method	T.	Clean	AA	Avg	NRR
PGD-AT	–	58.97	22.84	40.91	32.93
TRADES	–	57.35	25.60	41.48	35.40
MART	–	57.16	22.43	39.80	32.22
ARD	nat.	59.52	23.05	41.29	33.25
RSLAD	nat.	60.48	21.63	41.06	31.87
AdaAD	nat.	61.66	25.62	43.64	36.20
LBGAT	nat.	59.63	28.55	44.09	38.61
ARREST	nat.	67.69	24.69	46.19	36.18
HAT	nat.	62.81	26.91	44.86	37.68
ADR	self	64.50	29.04	46.77	40.05
CFA (ours)	nat.	65.04	30.14	47.59	41.19

F EXPERIMENTAL DETAILS

Table 22: Natural teacher checkpoints used by the shipped models. The teachers are trained only on clean examples and have zero adversarial accuracy.

Dataset	Architecture	Epochs	Clean
CIFAR-10	ResNet-18	200	95.32
CIFAR-100	ResNet-18	200	77.66
Tiny-ImageNet	ResNet-18	200	66.29

Every number we report is the **final** epoch, never a checkpoint selected on a robustness metric. This matters when reading Table 29: RPAT selects on PGD-20 and ADR on PGD-10, so those rows are upper envelopes of a training run where ours is its endpoint.

Teacher. One naturally trained network per dataset, no adversarial examples anywhere in its training, and the student is warm-started from the same checkpoint. ResNet-18, SGD at learning rate 0.1 under a one-cycle schedule, batch size 128, no augmentation beyond the standard crop and flip, for 200 epochs on each dataset. The resulting teachers are summarized in Table 22. The teacher’s own robustness is 0 — it is never attacked and is not meant to survive attack; Figure 3 is the ablation over how long it is trained.

Student. The main ResNet-18 configurations are listed below. Unless noted otherwise, CIFAR-100 ablations use 100 epochs at $8.8/255$, with a 10-epoch AWP warmup.

	CIFAR-10	CIFAR-100	Tiny-ImageNet-200
Backbone	ResNet-18	ResNet-18	ResNet-18
Optimizer	AdamW	AdamW	AdamW
Learning rate / schedule	0.021, one-cycle	0.021, one-cycle	0.021, one-cycle
Epochs	150	150	100
Batch size	128	128	128
Training radius ϵ_{train}	8/255	8/255	8.8/255
Inner attack	PGD-10, step 2/255	PGD-10, step 2/255	PGD-10, step 2/255
Allocation exponent p	1	1	1
Radius box $[w_{\text{lo}}, w_{\text{hi}}]$	[0.5, 1.5]	[0.5, 1.5]	[0.5, 1.5]
Classifier head	frozen	frozen	frozen
Weight averaging (start, decay)	0.2, 0.999	0.2, 0.999	0.2, 0.999
AWP (γ , warmup)	0.005, 15 ep	0.005, 15 ep	0.005, 10 ep
Teacher checkpoint	clean, 200 ep	clean, 200 ep	clean, 200 ep

The head being frozen is what removes τ and β from the method: with the classifier inherited and never updated, the head-KD term of earlier drafts has no parameters to reach and is switched off entirely. p is not tuned. The training-radius sweep is reported in Table 17; evaluation is at 8/255 in every cell.

Evaluation. We report AutoAttack accuracy at ℓ_∞ radius 8/255 on the full test set; NRR is its harmonic mean with clean accuracy. Additional attacks are in Appendices I and K. Ported baselines retain their published recipes unless stated otherwise.

Repeated training. Each reported row is an individual run. For the 100-epoch CIFAR-100 configuration at 8.8/255, seed 1 gives 62.00/32.49/30.82/28.74 clean/PGD-20/CW/AA, compared with 62.17/32.37/30.93/28.86 for seed 0. These two runs provide a limited reproducibility check, not a variance estimate or a significance threshold for other comparisons. In particular, small differences between regression targets or nearby operating points require further repetitions.

Compute, teacher included. A single RTX 5080. One student cell is about 2.2 hours on CIFAR-100 at 100 epochs, of which the AutoAttack evaluation is roughly 7 minutes; Tiny-ImageNet is longer in proportion to its test set.

The teacher has to be counted, since calling it free would otherwise hide a training run. It is not free, but it is cheap: a clean epoch has no inner maximization, so it costs 6.8 seconds against 74.5 for an adversarial one, and 200 clean epochs come to 0.39 hours. The teacher is therefore 19% of one student run and is amortized over every cell that reads it, the teacher ladder of Figure 3 included. Table 23 puts the total against the baselines measured on the same machine.

Table 23: Wall-clock training cost on one RTX 5080, CIFAR-100, ResNet-18, as logged by the runs reported in this paper. Evaluation is excluded. Adversarial distillation additionally requires an adversarially trained teacher, which is not counted in its row and is the more expensive artifact.

Method	Epochs	Teacher (h)	Student (h)	Total (h)
Natural teacher (ours, reused)	200 clean	–	0.39	0.39
CFA (ours)	100 adv.	0.39	2.07	2.46
Label CE at our regime	100 adv.	0.39	3.22	3.61
PGD-AT, initialized at the teacher	100 adv.	0.39	1.72	2.11
ARD	100 adv.	0.39	1.52	1.91
RSLAD	100 adv.	0.39	1.71	2.10
AdaAD	100 adv.	0.39	2.79	3.18
LBGAT	100 adv.	joint	1.88	1.88
HAT	50 adv.	0.39	1.10	1.49
ADR	200 adv.	self	3.52	3.52

Two readings follow. The anchor is not bought with compute, since it costs less end to end than label cross-entropy at the same regime and about what PGD-AT costs once the teacher is included. And the comparison with adversarial distillation is favourable in a way the table understates, because

those rows assume a robust teacher that the current literature obtains from a WideResNet trained on generated data, which is more expensive than every row here combined.

G ADDITIONAL TEACHER-CHECKPOINT RESULTS

Figure 3 shows one natural-training trajectory; Table 24 reports an independently trained trajectory. Each listed teacher was trained to completion under its own one-cycle schedule, then used to initialize a 50-epoch student with the same training procedure. These experiments vary the teacher checkpoint, not just the target features.

Table 24: Independent CIFAR-100 teacher trajectory (teacher seed 1; 50-epoch students with seed 0). Teacher metrics are measured on held-out data.

Epoch	Teacher			Student		
	Clean	Margin	Sensitivity	Clean	AA	NRR
20	73.87	3.074	3.911	63.82	23.22	34.05
40	75.57	3.931	4.542	64.36	23.86	34.81
100	77.27	4.506	4.678	63.68	25.61	36.53
150	77.68	4.495	4.262	63.37	25.49	36.36
200	77.90	4.542	4.129	63.02	26.16	36.97
300	78.27	4.523	3.835	62.34	25.94	36.64

The second trajectory again moves from higher student clean accuracy at earlier checkpoints to higher AA at later ones. The strongest AA occurs at 200 rather than 300 epochs, although the difference is small. Feature sensitivity has only moderate rank correlation with student clean accuracy on these six checkpoints (Spearman 0.54, versus 0.83 on the same grid of the first trajectory). The matched teachers in Table 2 further show that teacher clean accuracy alone does not determine student clean accuracy. None of these measurements supplies a validated teacher-only selection rule.

H THE RECIPE CONTROLS

Unless explicitly ablated, CFA warm-starts from the natural teacher and trains with AdamW at 0.021 under a one-cycle schedule, while the baselines retain their authors’ recipes. Table 25 asks whether that recipe is what wins: it runs our own objective, and label cross-entropy for reference, under both recipes and both initializations. Every row is measured here at the base regime, which carries no weight averaging, no AWP and a uniform radius, on CIFAR-100 with ResNet-18 for 100 epochs.

Table 25: The recipe controls. *Ours* is AdamW at 0.021 under a one-cycle schedule with the teacher warm start; *theirs* is SGD at 0.1 decayed by ten at epochs 70 and 90, the schedule PGD-AT uses here. The training radius and every other field are held fixed within each pair.

Objective	Recipe	Clean	AA	Avg	NRR
Anchor	ours, teacher init	61.29	25.36	43.33	35.88
Anchor	ours, random init	60.81	25.21	43.01	35.66
Anchor	theirs, teacher init	59.65	25.54	42.60	35.77
Label cross-entropy	ours, teacher init	54.60	19.45	37.03	28.68
Label cross-entropy	theirs, teacher init	56.66	20.91	38.79	30.55

Changing initialization or schedule changes the anchor’s NRR by 0.22 and 0.11, respectively. Cross-entropy improves by 1.87 NRR under its own schedule, so Table 5 includes that stronger baseline. The anchor remains ahead under both tested schedules. These comparisons support an objective-level contribution within the tested recipes, without establishing significance from single runs or claiming exhaustive tuning of the alternatives.

Table 26: Successive additions to the CIFAR-100 ResNet-18 anchor (100 epochs, $\epsilon_{\text{train}} = 8.8/255$).

	Clean	AA	NRR
Anchor ($p = 0$)	61.29	25.36	35.88
+ sensitivity-matched ϵ	63.09	25.74	36.56
+ weight averaging	62.30	27.69	38.34
+ AWP	62.17	28.86	39.42

I GRADIENT MASKING

We test for gradient masking with AutoAttack, independent attacks, and PGD sweeps (Athalye et al., 2018). Table 27 reports these checks for a 100-epoch CIFAR-100 repeat with 62.00% clean accuracy.

Table 27: Gradient-masking checks on all 10,000 CIFAR-100 test images at $8/255$. Individual attacks are evaluated independently.

Check	Setting	Accuracy
<i>white box against black box</i>		
APGD-CE	white box, untargeted	31.80
APGD-T	white box, targeted	28.83
FAB-T	white box, boundary	29.16
Square	black box , 5000 queries	34.82
AutoAttack	standard evaluation	28.73
<i>radius sweep, PGD-20</i>		
$\epsilon = 2/255$		54.62
$\epsilon = 4/255$		47.03
$\epsilon = 8/255$		32.46
$\epsilon = 12/255$		20.88
$\epsilon = 16/255$		11.94
$\epsilon = 32/255$		1.05
$\epsilon = 64/255$		0.00
<i>step sweep, $\epsilon = 8/255$</i>		
PGD-10 / 20 / 50 / 100		33.17 / 32.57 / 32.30 / 32.24
<i>transfer</i>		
Natural teacher $\rightarrow$ student	PGD-20 built on the teacher	60.42

AutoAttack yields 28.73% accuracy. PGD accuracy falls monotonically with attack radius, reaching zero at $64/255$, and decreases with additional steps. These checks show no indication of gradient masking under the evaluated attacks.

The transfer row is worth reading separately. The student inherits the teacher’s classifier and is initialized at the teacher’s features, so adversarial examples built on the teacher might be expected to carry over; they cost the student 1.58 points against its clean accuracy. Sharing a head is therefore not sharing a vulnerability, which is the same conclusion Section 5.4 reaches from class separation: what the anchored student changes is the position of its features, and the teacher’s own adversarial directions do not point at the student’s boundaries.

J NOTES ON THE BASELINES

Several rows of Table 3 run a specific variant of the method they name, or are grouped in a way that needs saying. This section is those decisions; the table carries none of them.

ARREST uses its published 20-epoch recipe in Table 6. The main table instead uses 150 epochs, AdamW, WA, and AWP to compare its objective with CFA under the longer training stack; AR-REST retains a uniform radius. Both versions start from the same natural teacher. We read its representation after global pooling, where our anchor and geometry measurements are defined.

LBGAT, and why its CIFAR-10 cell is empty. The authors release a training script for CIFAR-100 only, and their CIFAR-10 entries are released as pretrained WideResNet checkpoints. Running their CIFAR-100 recipe on CIFAR-10 diverges here within thirty steps: the squared error is reduced with mean over batch and classes, so a ten-class problem receives ten times the gradient of a hundred-class one at equal logit error, and $\text{lr} = 0.1$ is above the stability threshold that follows. Lowering the learning rate trains (at 0.01 the natural branch reaches 49% within 180 steps), but a learning rate we chose is not their recipe, so the cell is left empty rather than filled with a number of our own construction. LBGAT trains its natural branch jointly with the robust one, so unlike the distillation rows its teacher is not obtained for free, and we group it accordingly. The two datasets run different variants because the authors publish different ones: CIFAR-100 is LBGAT6 and CIFAR-10 is LBGAT0, their release lists CIFAR-10 checkpoints under LBGAT0 alone, and $\beta = 6$ is their CIFAR-100 setting. We report the configuration each dataset has a published counterpart for rather than forcing one β across both.

HAT uses a naturally trained network only to *label* helper points beyond the ball, never as a target, so it belongs with the trade-off methods and not the distillation ones. It is run on its own one-cycle schedule rather than ours, which collapses it to chance on both datasets — the same allowance we make for LBGAT. On CIFAR-10 the port reproduces HAT’s published no-extra-data figure to within 0.31 clean and 0.27 AutoAttack (84.90/49.08 reported, ResNet-18), which is what licenses reporting its CIFAR-100 row at all: HAT publishes no CIFAR-100 result without additional data, its 61.50/28.88 being trained with DDPM-generated samples, so that column has no published figure to be read against.

ADR is run on separate hardware and its row is filled from there. Two things have to hold when it is. ADR reports two configurations — AT+ADR at 56.10/26.87 on CIFAR-100 and AT+WA+AWP+ADR at 57.36/28.50 — and the row has to name which, the second being the one component-matched to our recipe, which carries both weight averaging and AWP. ADR is also self-distillation: its teacher is an exponential moving average of the student, no external network is read, and the student trains from scratch, so its row inherits nothing of our natural teacher.

What the published tables are blocked by. Table 29 and Table 30 separate their rows by network, and the separations are not cosmetic. RPAT’s “ResNet-18” is `ResNet(Bottleneck, [2, 2, 2, 2])` in its released code, 14.35M parameters against the 11.27M of the standard `BasicBlock` network used by us and by ADR; they also ship a `pre_resnet18` at 11.27M and use `PreActResNet-18` explicitly elsewhere, so the distinction is theirs. HAT’s Tiny-ImageNet row is `PreActResNet-18`: ADR lists it under ResNet-18, but the figures are quoted from HAT’s own Table 9, whose caption says `PreAct`, and HAT’s code refuses anything else on that dataset — same parameter count as ours, different network, and its two companions from that table are shown beside it. F²AT’s plain SAT baseline reaches 31.52/9.37 where ADR’s plain AT reaches 45.87/18.06, a gap no difference of method explains, so its rows are separated too. TE’s architecture we could not verify — it is ADR’s attribution to a paper we do not have — and it is marked accordingly. ADR trains Tiny-ImageNet for 80 epochs against our 100, which favours us and is stated here rather than left implicit. LBGAT reports that dataset without AutoAttack and B-MTARD reports it on MobileNet-v2, so neither appears; CURE, ARREST and Generalist++ do not report it at all.

IGDM is reported at $\alpha = 1$. Its published $\alpha = 20$ diverges to chance accuracy on both datasets when the teacher is natural, as documented in Table 19.

Teacher-only KL with the training stack. This diagnostic combines probability-target distillation with WA and AWP; it is not a reproduction of a published method. Temperature is applied only to the teacher, and the best tested stacked setting, $\tau = 4$, gives 56.97/26.96 clean/AA; $\tau = 16$ gives 42.45/20.10. The comparison concerns this KL training design, not all logit-based supervision. In particular, fixed-head logit MSE is evaluated separately in Table 13, where it remains competitive with feature regression.

K PER-ATTACK METRICS

Table 28 gives every attack we measure for our own cells. The main tables report clean accuracy and AutoAttack because those are the two axes the trade-off is defined on, and because PGD-20 is not a faithful reading of this method.

Table 28: Per-attack accuracy (%) for our cells. ResNet-18, $\epsilon = 8/255$, final-epoch weight-averaged model, full test set.

Dataset	Cell	Clean	FGSM	PGD-20	CW	AA	NRR
CIFAR-100	CFA, head frozen (shipped)	62.17	36.91	32.37	30.93	28.86	39.42
	the same cell at seed 1	62.00	36.79	32.49	30.82	28.74	39.27
	ℓ_1 gradient norm	62.16	36.48	31.97	30.56	28.57	39.15
	CFA, head-KD term retained	62.35	38.80	36.26	30.65	28.68	39.29
	CFA, $\epsilon_{tr} = 6/255$	68.07	35.88	28.40	28.16	25.82	37.44
	CFA, $\epsilon_{tr} = 7/255$	65.61	36.32	30.34	29.64	27.44	38.70
	CFA, $\epsilon_{tr} = 8/255$	63.89	36.67	31.41	29.83	27.78	38.72
	CFA, $\epsilon_{tr} = 10/255$	59.99	36.50	32.82	30.53	28.77	38.89
	CFA, no WA / AWP	61.29	33.97	28.23	27.19	25.36	35.88
	Logit anchor, our stack	56.97	35.82	33.68	28.94	26.96	36.60
	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
CIFAR-10	CFA, $\epsilon_{tr} = 8.8/255$ (shipped)	84.96	59.58	52.98	53.94	51.74	64.31
	CFA, $\epsilon_{tr} = 6/255$	89.64	59.37	49.29	50.20	47.88	62.42
	CFA, $\epsilon_{tr} = 7/255$	88.30	59.59	51.43	52.64	50.27	64.07
	CFA, $\epsilon_{tr} = 8/255$	87.22	59.92	52.43	53.55	51.15	64.48
	CFA, $\epsilon_{tr} = 10/255$	83.29	58.98	53.25	53.88	51.79	63.87
	RPAT++ (our reproduction)	82.41	59.73	55.72	52.88	50.75	62.82

Why PGD-20 is not the metric here. The student inherits the teacher’s classifier and never retrains it. PGD-20 ascends the cross-entropy *through that classifier*, so it reads a head that was fitted for a different feature distribution; AutoAttack, which includes a targeted APGD, and CW_∞ , which attacks the logit margin directly, do not depend on that fit in the same way. The consequence is measurable on a controlled pair. The first two rows of Table 28 share a backbone objective, a schedule and an attack, and differ only in whether the classifier is refit:

	PGD-20	CW	AA
head-KD term retained	36.26	30.65	28.68
head frozen	32.63	30.66	28.77
difference	−3.63	+0.01	+0.09

Changing the classifier moves PGD-20 by 3.63 points and moves CW and AutoAttack by hundredths. The same pattern recurs across our ablations—the head-temperature sweep, the label-smoothing sweep, the directional-versus-raw comparison, and the radius sweep all show PGD-20 moving against CW and AutoAttack. We therefore report PGD-20 for completeness and read the trade-off on clean accuracy and AutoAttack, which is also what ADR and ARREST report in their main tables.

L PUBLISHED RESULTS, FOR CONTEXT

Table 3 contains only numbers we measured ourselves. Table 29 collects the published figures for the same and neighbouring methods, so that our runs can be placed against the literature without the two being mixed in one column. These are quoted as reported and were produced by other codebases, other schedules and other checkpoint-selection rules—RPAT selects its best checkpoint by PGD-20 and ADR by PGD-10, where every number of ours is the final epoch—so they should be compared within a source rather than across.

Read against Table 3 and restricted to the ResNet-18 blocks, which are the ones our numbers are comparable with, our CFA row exceeds every published Avg and NRR on both datasets: 64.75/27.98 at Avg 46.37 on CIFAR-100 and 87.37/51.33 at Avg 69.35 on CIFAR-10.

Table 29: Published results, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. **The architecture changes between blocks** and is named in each heading; only the ResNet-18 blocks are comparable with Table 3. These were produced by other codebases, schedules and checkpoint-selection rules, so they should be compared within a source rather than across.

Method	Teacher	Clean	AA	Avg	NRR	Source
<i>CIFAR-100, ResNet-18</i>						
PGD-AT	–	56.56	25.02	40.79	34.69	RPAT
TRADES	–	55.39	24.51	39.95	33.98	RPAT
MART	–	49.83	25.00	37.41	33.30	RPAT
Consistency-AT	–	58.53	25.39	41.96	35.42	RPAT
Consistency-AT + RPAT	–	60.33	26.31	43.32	36.64	RPAT
F ² AT	–	54.19	23.24	38.71	32.53	F ² AT
Generalist++	co-tr.	62.97	23.96	43.47	34.71	Generalist++
ADR (AT + ADR)	self	56.10	26.87	41.48	36.34	ADR
ADR (AT + WA + AWP + ADR)	self	57.36	28.50	42.93	38.08	ADR
<i>CIFAR-10, ResNet-18</i>						
PGD-AT	–	82.78	44.63	63.71	57.99	CURE
TRADES	–	82.41	48.37	65.39	60.96	CURE
MART	–	80.70	47.49	64.10	59.79	CURE
ST-AT	–	83.10	50.50	66.80	62.82	CURE
IAD	robust	80.63	50.17	65.40	61.85	CURE
Consistency-AT + RPAT	–	84.12	48.98	66.55	61.91	RPAT
Generalist++	co-tr.	89.09	46.07	67.58	60.73	Generalist++
ARREST	natural	86.63	46.14	66.39	60.21	ARREST
CURE	self	86.76	49.69	68.23	63.19	CURE
ADR (AT + ADR)	self	82.41	50.38	66.40	62.53	ADR
<i>CIFAR-10, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	83.50	49.89	66.70	62.46	RPAT
AWP (Wu et al., 2020)	–	81.11	50.09	65.60	61.93	RPAT
KD + SWA (Chen et al., 2021)	robust	84.06	49.82	66.94	62.56	RPAT
TE (Dong et al., 2022)	–	82.04	50.12	66.08	62.23	RPAT
ReBAT (Wang et al., 2023)	–	82.09	50.72	66.41	62.70	RPAT
RPAT++ (Liu et al., 2025)	–	82.63	51.00	66.82	63.07	RPAT
<i>CIFAR-100, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	57.26	25.83	41.55	35.60	RPAT
AWP (Wu et al., 2020)	–	54.10	25.16	39.63	34.35	RPAT
KD + SWA (Chen et al., 2021)	robust	57.17	25.66	41.42	35.42	RPAT
TE (Dong et al., 2022)	–	56.41	25.84	41.13	35.44	RPAT
ReBAT (Wang et al., 2023)	–	56.13	27.60	41.87	37.00	RPAT
RPAT++ (Liu et al., 2025)	–	56.84	27.68	42.26	37.23	RPAT
<i>CIFAR-10, WideResNet-34-10</i>						
WA (Izmailov et al., 2018)	–	87.66	52.65	70.16	65.79	RPAT
MMA (Ding et al., 2020)	–	87.80	43.10	65.45	57.82	RPAT
AWP (Wu et al., 2020)	–	85.63	53.32	69.48	65.72	RPAT
KD + SWA (Chen et al., 2021)	robust	87.45	53.59	70.52	66.46	RPAT
TE (Dong et al., 2022)	–	85.97	52.88	69.43	65.48	RPAT
ReBAT (Wang et al., 2023)	–	85.25	54.78	70.02	66.70	RPAT
ADR (Wu et al., 2024)	self	84.67	53.25	68.96	65.38	RPAT
CURE (Gowda et al., 2024)	self	87.05	52.10	69.58	65.19	RPAT
RPAT++ (Liu et al., 2025)	–	86.76	54.97	70.87	67.30	RPAT

The restriction is not a convenience. The WideResNet-34-10 block exceeds ours on both criteria—RPAT++ reaches Avg 70.87 and NRR 67.30 there—because a 48.3M network is a different model, not a better method, and Tables 4 and 21 are where that comparison belongs. The same caution applies to the PreActResNet-18 blocks in the other direction: RPAT++ falls to 82.63/51.00 there, and reading that as a defeat for RPAT would be the identical error. We state all of this as a comparison across codebases and architectures rather than as a controlled result, which is what Table 3 is for.

Table 30: Published Tiny-ImageNet-200 results, ResNet-18, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. Our row is above all of them on both axes. The blocks are separated because the networks are not the same, and Appendix J says how each differs.

Method	Clean	AA	Avg	NRR	Source
CFA (ours)	55.16	20.54	37.85	29.93	—
AT + WA + ADR	48.55	20.23	34.39	28.56	ADR
AT + WA + AWP + ADR	48.27	20.12	34.20	28.40	ADR
TRADES + WA + AWP + ADR	51.38	19.48	35.43	28.25	ADR
TRADES + WA + ADR	51.99	19.17	35.58	28.01	ADR
AT + WA + AWP	48.61	19.58	34.09	27.92	ADR
AT + WA	49.10	19.30	34.20	27.71	ADR
TE*	47.46	18.29	32.88	26.40	ADR
AT	45.87	18.06	31.96	25.92	ADR
TRADES	48.49	17.35	32.92	25.56	ADR
<i>PreActResNet-18 — same parameter count, different network</i>					
HAT	52.60	18.14	35.37	26.98	HAT
TRADES	48.25	17.17	32.71	25.35	HAT
AT	47.76	17.92	32.84	26.02	HAT
<i>Bottleneck ResNet-18 (14.35M) and an unstated setup — see caption</i>					
Consistency-AT + RPAT	49.74	18.84	34.29	27.33	RPAT
PGD-AT + RPAT	47.68	17.77	32.73	25.89	RPAT
Consistency-AT	46.54	17.60	32.07	25.54	RPAT
MART	39.70	17.18	28.44	23.98	RPAT
F ² AT	40.54	13.13	26.84	19.84	F ² AT
SEAT	35.51	11.37	23.44	17.22	F ² AT
SAT	31.52	9.37	20.45	14.45	F ² AT

M THE CURE REPRODUCTION

Seven runs of the official CURE repository, from its own code and our shared CIFAR data, none of which reproduce the published 86.76/49.69 on CIFAR-10. They fall short on *different* axes: the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Running the repository unmodified gives 65.03/29.74. Computing its RGP prominence mask on absolute values raises AA by 15.9 points (29.74 $\rightarrow$ 45.63), which localizes the discrepancy to that step without closing it.

We re-evaluated all five surviving checkpoints with our own pipeline and it agrees with the repository’s own evaluation to within 0.05 AutoAttack on every one, so the gap is in training rather than in measurement. For contrast, RPAT++ reproduces to within 0.31 AutoAttack of its published numbers on both datasets under the same conditions. The CURE row of Table 29 is therefore the paper’s own number, and CURE is omitted from Table 3.

N A CORRECTION TO THE AWP PROXY, AND WHAT IT MOVED

AWP ascends an auxiliary objective in weight space before each descent step. In our implementation the closure used for that ascent had drifted from the closure used for descent, in two ways. It retained a head-distillation term that `freeze_head` removes from training, so AWP was maximizing a quantity partly composed of a term the model never optimizes; and a detach intended for the normalized feature was applied on a path that feeds the unnormalized one. Measured on CIFAR-100 before the fix, the spurious term was 7.15 against the anchor’s 4.15 and contributed 40.8% of the

AWP backbone gradient, at cosine 0.918 to the correct direction. Separately, the head was frozen after the proxy was copied, which is harmless only because the proxy’s first step is delayed by the AWP warmup.

We report this because it is the kind of defect that changes a paper’s numbers, and because here it did not. Re-running the shipped recipe with both closures agreeing gives 62.17/28.86 against 62.65/28.77 on CIFAR-100 and 84.96/51.74 against 85.58/51.79 on CIFAR-10: NRR moves by 0.01 and 0.22, no ordering in any table changes, and clean accuracy moves more than robustness does. AWP is evidently insensitive to a large, well-aligned perturbation of its ascent direction, which is consistent with its role here as a regularizer of sharpness rather than a precisely targeted objective. All numbers reported for the shipped recipe are post-correction. Two exceptions are stated where they appear: the head-freeze comparison in Appendix K and the ϵ_{train} sweep in Table 17 were measured before it. In each, every arm shares the defect, so the comparison is internally consistent even though its absolute values differ from the corrected ones by a few hundredths.

REFERENCES

- Anish Athalye, Nicholas Carlini, and David Wagner. Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples. In *International conference on machine learning*, pp. 274–283. PMLR, 2018.
- Yogesh Balaji, Tom Goldstein, and Judy Hoffman. Instance adaptive adversarial training. In *arXiv*, 2019.
- Tianlong Chen, Zhenyu Zhang, Sijia Liu, Shiyu Chang, and Zhangyang Wang. Robust overfitting may be mitigated by properly learned smoothening. In *International Conference on Learning Representations*, 2021.
- Minhao Cheng, Qi Lei, Pin-Yu Chen, Inderjit Dhillon, and Cho-Jui Hsieh. Customized adversarial training for improved robustness. In *IJCAI*, 2022.
- Seungju Cho and Changick Kim. Direction-preserving fast adversarial training. In *under review*, 2026.
- Jiequan Cui, Shu Liu, Liwei Wang, and Jiaya Jia. Learnable boundary guided adversarial training. In *ICCV*, 2021.
- Gavin Weiguang Ding, Yash Sharma, Kry Yik Chau Lui, and Ruitong Huang. MMA training: Direct input space margin maximization through adversarial training. In *ICLR*, 2020.
- Yinpeng Dong, Ke Xu, Xiao Yang, Tianyu Pang, Zhijie Deng, Hang Su, and Jun Zhu. Exploring memorization in adversarial training. In *International Conference on Learning Representations*, 2022.
- Micah Goldblum, Liam Fowl, Soheil Feizi, and Tom Goldstein. Adversarially robust distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pp. 3996–4003, 2020.
- Shruthi Gowda, Bahram Zonooz, and Elahe Arani. Conserve-update-revise to cure generalization and robustness trade-off in adversarial training. In *ICLR*, 2024.
- Bo Huang, Mingyang Chen, Yi Wang, Junda Lu, Minhao Cheng, and Wei Wang. Boosting accuracy and robustness of student models via adaptive adversarial distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 24668–24677, 2023.
- Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Conference on Uncertainty in Artificial Intelligence*, 2018.
- Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009.
- Ya Le and Xuan Yang. Tiny imagenet visual recognition challenge. *CS 231N*, 7(7):3, 2015.

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- Hongsin Lee, Seungju Cho, and Changick Kim. Indirect gradient matching for adversarial robust distillation. In *ICLR*, 2025.
- Yanyun Liu et al. Failure cases are better learned but boundary says sorry: Facilitating smooth perception change for accuracy-robustness trade-off in adversarial training. In *ICCV*, 2025.
- Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. *arXiv preprint arXiv:1706.06083*, 2017.
- Tianyu Pang, Xiao Yang, Yinpeng Dong, Hang Su, and Jun Zhu. Bag of tricks for adversarial training. *arXiv preprint arXiv:2010.00467*, 2020.
- Rahul Rade and Seyed-Mohsen Moosavi-Dezfooli. Reducing excessive margin to achieve a better accuracy vs. robustness trade-off. In *International Conference on Learning Representations*, 2022.
- Leslie Rice, Eric Wong, and J. Zico Kolter. Overfitting in adversarially robust deep learning. In *ICML*, 2020.
- Satoshi Suzuki, Shin’ya Yamaguchi, Shoichiro Takeda, Sekitoshi Kanai, Naoki Makishima, Atsushi Ando, and Ryo Masumura. ARREST: Adversarial finetuning with latent representation constraint to mitigate the accuracy-robustness tradeoff. In *ICCV*, 2023.
- Jihoon Tack, Sihyun Yu, Jongheon Jeong, Minseon Kim, Sung Ju Hwang, and Jinwoo Shin. Consistency regularization for adversarial robustness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 8414–8422, 2022.
- Dimitris Tsipras, Shibani Santurkar, Logan Engstrom, Alexander Turner, and Aleksander Madry. Robustness may be at odds with accuracy. *arXiv preprint arXiv:1805.12152*, 2018.
- Yisen Wang, Difan Zou, Jinfeng Yi, James Bailey, Xingjun Ma, and Quanquan Gu. Improving adversarial robustness requires revisiting misclassified examples. In *International Conference on Learning Representations*, 2020.
- Yisen Wang et al. Robust weight perturbation for adversarial training. In *NeurIPS*, 2023.
- Dongxian Wu, Shu-Tao Xia, and Yisen Wang. Adversarial weight perturbation helps robust generalization. *Advances in Neural Information Processing Systems*, 33:2958–2969, 2020.
- Yu-Yu Wu, Hung-Jui Wang, and Shang-Tse Chen. Annealing self-distillation rectification improves adversarial training. In *ICLR*, 2024.
- Hongyang Zhang, Yaodong Yu, Jiantao Jiao, Eric Xing, Laurent El Ghaoui, and Michael Jordan. Theoretically principled trade-off between robustness and accuracy. In *International conference on machine learning*, pp. 7472–7482. PMLR, 2019.
- Shiji Zhao, Xizhe Wang, and Xingxing Wei. Mitigating accuracy-robustness trade-off via balanced multi-teacher adversarial distillation. *IEEE TPAMI*, 2024.
- Jianing Zhu, Jiangchao Yao, Bo Han, Jingfeng Zhang, Tongliang Liu, Gang Niu, Jingren Zhou, Jianliang Xu, and Hongxia Yang. Reliable adversarial distillation with unreliable teachers. *arXiv preprint arXiv:2106.04928*, 2021.
- Bojia Zi, Shihao Zhao, Xingjun Ma, and Yu-Gang Jiang. Revisiting adversarial robustness distillation: Robust soft labels make student better. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 16443–16452, 2021.